

POSTNIKOV TRUNCATION OF SIMPLICIAL OBJECTS IN AN ∞ -TOPOS AS A JOIN OF LOCALIZATIONS

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ABSTRACT. Let $\mathcal{E}$ be an ∞ -topos and $\mathcal{P} = \text{Fun}(\Delta^{\text{op}}, \mathcal{E})$ its ∞ -category of simplicial objects. Three classes of maps of $\mathcal{P}$ are canonical: the maps inverted by geometric realization $|-| = \text{colim}$, the maps inverted by the $(n+1)$ -coskeleton, and, for a left-exact localization L of $\mathcal{E}$, the maps inverted by L levelwise. Each is the class of equivalences of a reflective localization of $\mathcal{P}$, and each is closed under composition and two-out-of-three. We prove that the smallest class closed under composition and two-out-of-three containing the first two is exactly the class of maps f with $\tau_{\leq n}|f|$ an equivalence, and that adding the third gives exactly the maps with $L\tau_{\leq n}|f|$ an equivalence, the n -truncated L -sheaf of the realization. The identities hold on the nose, not merely after saturation, which answers a question left open in earlier work. The proof rests on one proposition: the geometric realization of the $(n+1)$ -coskeleton of a constant simplicial object cK is the Postnikov section $\tau_{\leq n}K$, with the coskeleton unit realizing to the truncation map. For spaces this is proved by a bisimplicial diagonal argument from the classical fact that the coskeleton of a Kan complex is its Postnikov section, and it passes to every ∞ -topos because coskeleta are finite limits and left-exact localizations commute with finite limits and with truncation.

1. INTRODUCTION

Throughout, $\mathcal{E}$ is an ∞ -topos, $\mathcal{P} := \text{Fun}(\Delta^{\text{op}}, \mathcal{E})$ is its ∞ -category of simplicial objects, $|-|: \mathcal{P} \rightarrow \mathcal{E}$ is the colimit functor (geometric realization), $c: \mathcal{E} \rightarrow \mathcal{P}$ is the constant functor, $\text{cosk}_{n+1}: \mathcal{P} \rightarrow \mathcal{P}$ is the $(n+1)$ -coskeleton, and $\tau_{\leq n}: \mathcal{E} \rightarrow \mathcal{E}$ is n -truncation. For a functor F we write W_F for the class of maps f such that Ff is an equivalence. The main result is the following identity between classes of maps of $\mathcal{P}$, where $\vee$ denotes the smallest class containing the equivalences, closed under composition and two-out-of-three, and containing both arguments.

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Theorem 1.1 (main theorem). *For every ∞ -topos $\mathcal{E}$, every $n \geq -1$ and every left-exact localization L of $\mathcal{E}$ applied levelwise to $\mathcal{P}$,*

$$W_{|-|} \vee W_{\text{cosk}_{n+1}} = W_{\tau_{\leq n}|-|}, \quad W_{|-|} \vee W_{\text{cosk}_{n+1}} \vee W_L = W_{L\tau_{\leq n}|-|}.$$

What is known. Classes of maps closed under composition and two-out-of-three, called *samenesses*, form a complete lattice $\text{Same}(C)$ whose joins are subtle [5]; the general *join theorem* (Theorem 2.2 below) is from [6], where the identity $W_{\text{weak equivalences}} \vee W_{\text{cosk}_{n+1}} = \{f : \pi_i f \text{ iso}, i \leq n\}$ was proved for simplicial *sets*, i.e. for $\mathcal{E} = \mathcal{S}$ and levelwise discrete objects, using Kan's Ex^∞ and the classical fact that the $(n+1)$ -coskeleton of a Kan complex is its n -th Postnikov section (Duskin [2]; [3, VI]). In [7] it was shown that in an ∞ -topos the join of a lex modality with the n -truncation is the n -truncated modality, and that in simplicial spaces ($\mathcal{E} = \mathcal{S}$) the *saturation* of $W_{|-|} \vee W_{\text{cosk}_{n+1}}$ is $W_{\tau_{\leq n}|-|}$, i.e. the two classes have the same local objects; whether the join itself is saturated was left open there. The facts about ∞ -topoi that we use — left-exact localizations commute with truncation, every ∞ -topos is a left-exact localization of a presheaf ∞ -category — are from [4], and the facts about bisimplicial sets from [3, IV.1].

What is new. The two identities of Theorem 1.1: the first closes the question of [7, §8] and generalizes the theorem of [6] from simplicial sets to simplicial objects in every ∞ -topos; the second adds the sheaf-theoretic sameness. Both follow from the join theorem once one knows that the unit $cK \rightarrow c\tau_{\leq n}K$ lies in the join, and this is Proposition 3.3: *the realization of the coskeleton unit $cK \rightarrow \text{cosk}_{n+1}cK$ is the truncation $K \rightarrow \tau_{\leq n}K$* . The proposition is not a formal consequence of the Kan-complex statement, since a constant simplicial space has non-discrete levels and its coskeleton has levels $K^{|\text{sk}_{n+1}\Delta^m|}$ that are far from truncated; it is proved by writing the coskeleton of a constant object as the diagonal of a bisimplicial set whose rows are coskeleta of Kan complexes. No result of this paper depends on a computation; a small computer check of Proposition 3.3 in the first non-trivial case accompanies the repository and is mentioned only in Remark 3.4.

Outline. Section 2 recalls *samenesses* and the join theorem. Section 3 collects the facts about realization and coskeleta and proves Proposition 3.3. Section 4 proves the first identity (Theorem 4.1) and Section 5 the second (Theorem 5.2).

2. SAMENESSES AND THE JOIN THEOREM

Definition 2.1. A *sameness* of an ∞ -category C is a class W of morphisms (closed under equivalence of morphisms in the arrow category) containing all equivalences, closed under composition, and satisfying two-out-of-three: if two of f , g , gf lie in W , so does the third. The *samenesses* form a complete lattice $\text{Same}(C)$ under inclusion; the meet is the intersection and the join $U \vee V$ is the smallest sameness containing $U \cup V$.

For a functor $F: C \rightarrow D$, the class $W_F := F^*(\text{Iso}) = \{f : Ff \text{ an equivalence}\}$ is a sameness, and more generally $F^*(W)$ is a sameness for every sameness W of D . For a sameness W of C let $\text{Loc}(W)$ be the class of W -local objects, those Z for which $\text{Map}(f, Z)$ is an equivalence for all $f \in W$, and let $\text{Sat}(W)$ be the *saturation*: the class of maps inverted by the localization $C \rightarrow C[W^{-1}]$; then $W \subseteq \text{Sat}(W)$, $\text{Loc}(W) = \text{Loc}(\text{Sat}(W))$, and $\text{Loc}(U \vee V) = \text{Loc}(U) \cap \text{Loc}(V)$ (Z is local for $U \vee V$ iff $U \vee V \subseteq \text{Map}(-, Z)^*(\text{Iso})$, a sameness, iff both U and V are contained in it). If $W = W_R$ for a reflector R onto a full subcategory, then W_R is saturated and $\text{Loc}(W_R)$ is the essential image of R ; and if $\text{Loc}(W)$ is a reflective subcategory with reflector R , then $\text{Sat}(W) = W_R$ [6, §2].

Theorem 2.2 (join theorem, [6, Thm. 4.1]). *Let $U, V, W \in \text{Same}(C)$ with $W \subseteq U \vee V$, and let $P: C \rightarrow C$ be a functor with a natural transformation $\pi: \text{id}_C \Rightarrow P$ such that (a) $\pi_X \in U \vee V$ for every X , and (b) $P(U) \subseteq W$ and $P(V) \subseteq W$. Then $U \vee V = P^*(W)$.*

Proof. $P^*(W)$ is a sameness containing U and V by (b), hence $U \vee V$. Conversely let $Pf \in W$. Naturality gives the square

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \pi_X \downarrow & & \downarrow \pi_Y \\ PX & \xrightarrow{Pf} & PY \end{array}$$

in which π_X, π_Y and Pf lie in $U \vee V$; two-out-of-three applied to $\pi_Y \circ f = Pf \circ \pi_X$ gives $f \in U \vee V$. $\square$

The theorem is stated for 1-categories in [6]; the proof uses only that the square commutes in the homotopy category, so it holds for ∞ -categories with the definition above.

3. REALIZATION, CONSTANTS AND COSKELETA

3.1. Realization and constants. Since Δ^{op} has an initial object, namely $[0]$ (which is terminal in Δ), its nerve is contractible and the colimit of a constant diagram is its value: $|cK| \simeq K$. Hence c is fully faithful and the unit $u_X: X \rightarrow c|X|$ of $|-| \dashv c$ satisfies $|u_X| \simeq \text{id}$, i.e. $u_X \in W_{|-|}$ for every X ; and a map $X \rightarrow cK$ factors uniquely as $c(g) \circ u_X$ with $g: |X| \rightarrow K$ its adjunct. Both $c|-|$ and cosk_{n+1} are reflectors (onto the constant and the $(n+1)$ -coskeletal objects), so $W_{|-|}$ and $W_{\text{cosk}_{n+1}}$ are saturated samenesses with these local objects.

3.2. Coskeleta are finite limits. Let $\Delta_{\leq k} \subseteq \Delta$ be the full subcategory on $[0], \dots, [k]$. The k -coskeleton is the right Kan extension along $\Delta_{\leq k}^{\text{op}} \hookrightarrow \Delta^{\text{op}}$ of the restriction, so

$$(\text{cosk}_k X)_m = \lim_{([j] \rightarrow [m]) \in (\Delta_{\leq k} / [m])^{\text{op}}} X_j,$$

a limit over a *finite* category (there are finitely many maps $[j] \rightarrow [m]$ with $j \leq k$), with unit $X \rightarrow \text{cosk}_k X$ an equivalence in levels $\leq k$. Equivalently, $(\text{cosk}_k X)_m$ is the cotensor $\text{Map}_{\mathcal{P}}(\text{sk}_k y[m], X)$ of X with the k -skeleton of the representable.

Lemma 3.1. *Let $F: \mathcal{E} \rightarrow \mathcal{E}'$ be a functor preserving finite limits. Then $F \circ \text{cosk}_k \simeq \text{cosk}_k \circ F$ as functors $\text{Fun}(\Delta^{\text{op}}, \mathcal{E}) \rightarrow \text{Fun}(\Delta^{\text{op}}, \mathcal{E}')$, compatibly with the units. In particular this holds for the inclusion $\iota: \mathcal{E} \hookrightarrow \text{Presh}(C)$ of an ∞ -topos into a presheaf ∞ -category and for its left-exact left adjoint L .*

Proof. Both sides are the finite limits displayed above, and F preserves them. $\square$

3.3. Constant objects. For $K \in \mathcal{E}$ and a space A write K^A for the cotensor. Then $(\text{cosk}_{n+1} cK)_m \simeq K^{|\text{sk}_{n+1} \Delta^m|}$, and the unit $cK \rightarrow \text{cosk}_{n+1} cK$ is, at level m , the map $K \rightarrow K^{|\text{sk}_{n+1} \Delta^m|}$ induced by $|\text{sk}_{n+1} \Delta^m| \rightarrow *$. The space $|\text{sk}_{n+1} \Delta^m|$ is contractible for $m \leq n+1$ and n -connected for $m \geq n+2$, with $|\text{sk}_{n+1} \Delta^{n+2}| = |\partial \Delta^{n+2}| \simeq S^{n+1}$; in particular the levels of $\text{cosk}_{n+1} cK$ are not truncated when K is not (for $K = S^1$ and $n = 0$, level m is $\text{Map}(K_{m+1}, S^1) \simeq S^1 \times \mathbb{Z}^{\binom{m}{2}}$, K_{m+1} the complete graph).

Lemma 3.2 ([7, Lemma 8.2]). *cK is $(n+1)$ -coskeletal iff K is n -truncated.*

Proof. If K is n -truncated then $K \rightarrow K^A$ is an equivalence for every n -connected space A (test against $\text{Map}(B, -): \text{Map}(A, \text{Map}(B, K)) \simeq \text{Map}(\tau_{\leq n} A, \text{Map}(B, K)) \simeq \text{Map}(B, K)$). Conversely, if cK is $(n+1)$ -coskeletal, the unit at level $n+2$, $K \rightarrow K^{S^{n+1}}$, is an equivalence, which says precisely that K is n -truncated. $\square$

The following proposition is the heart of the paper. Although the levels of $\text{cosk}_{n+1} cK$ are not truncated, its realization is.

Proposition 3.3 (coskeleta of constants realize to Postnikov sections). *Let $\mathcal{E}$ be an ∞ -topos, $K \in \mathcal{E}$ and $n \geq -1$. The realization of the coskeleton unit,*

$$K \simeq |cK| \longrightarrow |\text{cosk}_{n+1} cK|,$$

is the truncation map $K \rightarrow \tau_{\leq n} K$; in particular $|\text{cosk}_{n+1} cK| \simeq \tau_{\leq n} K$, naturally in K .

Proof. The case $\mathcal{E} = \mathcal{S}$. Choose a Kan complex K' with $|K'| \simeq K$. For a simplicial set A and a Kan complex Z write Z^A for the simplicial mapping space, $(Z^A)_k = \text{Hom}(A \times \Delta^k, Z)$, a Kan complex whose homotopy type is $\text{Map}(|A|, |Z|)$ [3, I.5, I.11]. Consider the three bisimplicial sets

$$\begin{aligned} S_{m,k} &= \text{Hom}(\Delta^m \times \Delta^k, K'), & T_{m,k} &= \text{Hom}(\text{sk}_{n+1} \Delta^m \times \Delta^k, K'), \\ T'_{m,k} &= \text{Hom}(\Delta^m \times \Delta^k, \text{cosk}_{n+1} K'), \end{aligned}$$

with the maps $S \rightarrow T$ (restriction along $\text{sk}_{n+1} \Delta^m \times \Delta^k \subseteq \Delta^m \times \Delta^k$) and $T \rightarrow T'$ (a map $\text{sk}_{n+1} \Delta^m \times \Delta^k \rightarrow K'$ restricts to $\text{sk}_{n+1}(\Delta^m \times \Delta^k) \subseteq$

$\mathrm{sk}_{n+1}\Delta^m \times \Delta^k$, i.e. gives a map $\Delta^m \times \Delta^k \rightarrow \mathrm{cosk}_{n+1}K'$. Both maps are natural in K' .

Columns (fixed m). $k \mapsto S_{m,k}$ is $K'^{\Delta^m} \simeq K'$, and $k \mapsto T_{m,k}$ is $K'^{\mathrm{sk}_{n+1}\Delta^m}$, of homotopy type $\mathrm{Map}(|\mathrm{sk}_{n+1}\Delta^m|, K) = (\mathrm{cosk}_{n+1}cK)_m$, the simplicial structure in m being induced by the coface and codegeneracy maps in both cases. Hence the simplicial spaces $m \mapsto |S_{m,\bullet}|$ and $m \mapsto |T_{m,\bullet}|$ are cK and $\mathrm{cosk}_{n+1}cK$, and $S \rightarrow T$ models the coskeleton unit. The realization of the simplicial space obtained by realizing a bisimplicial set in one direction is its diagonal [3, IV.1.7 and IV.1.9], so the realization of the unit is $\mathrm{diag}S \rightarrow \mathrm{diag}T$.

Rows (fixed k). By the exponential law, $m \mapsto S_{m,k}$ is the Kan complex K'^{Δ^k} , $m \mapsto T_{m,k}$ is $\mathrm{Hom}(\mathrm{sk}_{n+1}\Delta^m, K'^{\Delta^k}) = (\mathrm{cosk}_{n+1}(K'^{\Delta^k}))_m$, and $m \mapsto T'_{m,k}$ is $(\mathrm{cosk}_{n+1}K')^{\Delta^k}$; the row maps are the coskeleton unit of K'^{Δ^k} and the canonical comparison $\mathrm{cosk}_{n+1}(K'^{\Delta^k}) \rightarrow (\mathrm{cosk}_{n+1}K')^{\Delta^k}$. By the Kan-complex statement [6, Lemma 7.1] (cf. [2]), for a Kan complex Z the unit $Z \rightarrow \mathrm{cosk}_{n+1}Z$ is a map of Kan complexes inducing isomorphisms on π_i for $i \leq n$ at all base points, with $\pi_i(\mathrm{cosk}_{n+1}Z) = 0$ for $i > n$ (for $n = -1$: cosk_0Z is empty or contractible). Apply this to $Z = K'^{\Delta^k}$ and to $Z = K'$: the row map $S_{\bullet,k} \rightarrow T_{\bullet,k}$ is an n -equivalence to an n -truncated Kan complex, and the composite $K'^{\Delta^k} \rightarrow \mathrm{cosk}_{n+1}(K'^{\Delta^k}) \rightarrow (\mathrm{cosk}_{n+1}K')^{\Delta^k}$ is the Δ^k -exponential of the unit of K' , again an n -equivalence to an n -truncated Kan complex (evaluation at a vertex identifies $Z^{\Delta^k} \simeq Z$ naturally). Hence $T_{\bullet,k} \rightarrow T'_{\bullet,k}$ induces isomorphisms on all homotopy groups between Kan complexes, i.e. is a weak equivalence, for every k .

Diagonals. A map of bisimplicial sets that is a weak equivalence on each row induces a weak equivalence of diagonals [3, IV.1.7]. Applied to $T \rightarrow T'$ and to the rowwise equivalences $S_{\bullet,k} = K'^{\Delta^k} \rightarrow K'$ and $T'_{\bullet,k} = (\mathrm{cosk}_{n+1}K')^{\Delta^k} \rightarrow \mathrm{cosk}_{n+1}K'$ (evaluation at the vertex 0 of Δ^k , which is compatible with the coskeleton units), this gives the commutative square

$$\begin{array}{ccccc} \mathrm{diag}S & \longrightarrow & \mathrm{diag}T & \xrightarrow{\simeq} & \mathrm{diag}T' \\ \simeq \downarrow & & & & \downarrow \simeq \\ K' & \xrightarrow{\quad \text{unit} \quad} & \mathrm{cosk}_{n+1}K' & & \end{array}$$

in which the bottom map is the coskeleton unit of the Kan complex K' , an n -equivalence to an n -truncated Kan complex, i.e. the Postnikov n -section $K \rightarrow \tau_{\leq n}K$. This proves the proposition for $\mathcal{E} = \mathcal{S}$, and the identification is natural in K because all maps are natural in K' and an objectwise equivalence between natural transformations is a natural equivalence.

Presheaf ∞ -topoi. For a small ∞ -category C and $\mathcal{E} = \mathrm{Presh}(C) = \mathrm{Fun}(C^{\mathrm{op}}, \mathcal{S})$, the functors c , cosk_{n+1} (a finite limit), $|-|$ (a colimit) and $\tau_{\leq n}$ are computed objectwise, i.e. are postcomposition with their $\mathcal{S}$ -versions. The natural map $\tau_{\leq n}K \rightarrow |\mathrm{cosk}_{n+1}cK|$ of $\mathcal{S}$ -valued functors, which is an

equivalence by the first case, therefore induces an objectwise equivalence for presheaves of spaces on any C .

General ∞ -topoi. Write $\mathcal{E} = L \text{Presh}(C)$ for a small ∞ -category C and an accessible left-exact localization L with fully faithful right adjoint ι [4, 6.1.0.6]. For $K \in \mathcal{E}$, the coskeleton $\text{cosk}_{n+1} cK$ is computed in $\text{Presh}(C)$ (Lemma 3.1 for ι), and the colimit in $\mathcal{E}$ is L of the colimit in $\text{Presh}(C)$, so

$$|\text{cosk}_{n+1} cK|_{\mathcal{E}} \simeq L |\text{cosk}_{n+1} c \iota K|_{\text{Presh}(C)} \simeq L \tau_{\leq n} \iota K \simeq \tau_{\leq n} L \iota K \simeq \tau_{\leq n} K$$

by the presheaf case and $L \tau_{\leq n} \simeq \tau_{\leq n} L$ [4, 5.5.6.28]; under these identifications the realized unit is L applied to the truncation map of ιK , which is the truncation map of K . $\square$

Remark 3.4. The first non-trivial instance is $n = 0$, $K = S^1$: the simplicial space $m \mapsto \text{Map}(|\text{sk}_1 \Delta^m|, S^1)$, whose levels are $S^1 \times \mathbb{Z}^{\binom{m}{2}}$, has contractible realization. Since $B\mathbb{Z}$ is a simplicial abelian group, the diagonal $\text{diag} T$ of the proof is one too, with m -simplices the normalized $\mathbb{Z}$ -valued 1-cocycles on $\text{sk}_1 \Delta^m \times \Delta^m$, and its homotopy groups are the homology of its Moore complex. A script in the repository computes the ranks of this Moore complex over $\mathbb{Q}$ through degree 4 (all rational homotopy groups $\pi_0, \dots, \pi_3$ vanish, while the same computation with Δ^m in place of $\text{sk}_1 \Delta^m$, which models S^1 itself, gives $\pi_1 \otimes \mathbb{Q} = \mathbb{Q}$). Nothing in the paper depends on it.

4. THE FIRST IDENTITY

Theorem 4.1. *Let $\mathcal{E}$ be an ∞ -topos and $n \geq -1$. In $\text{Same}(\text{Fun}(\Delta^{\text{op}}, \mathcal{E}))$,*

$$W_{|-|} \vee W_{\text{cosk}_{n+1}} = W_{\tau_{\leq n}|-|},$$

and the local objects of this class are the constant simplicial objects on n -truncated objects of $\mathcal{E}$.

Proof. Apply the join theorem with $U = W_{|-|}$, $V = W_{\text{cosk}_{n+1}}$, $W = \text{Iso}$ and $P = c\tau_{\leq n}|-|$, with π_X the composite $X \xrightarrow{u_X} c|X| \xrightarrow{t_{|X|}} c\tau_{\leq n}|X|$, where $t_K := c(K \rightarrow \tau_{\leq n} K)$. Condition (b): P inverts $W_{|-|}$ trivially, and it inverts $W_{\text{cosk}_{n+1}}$ because $W_{\text{cosk}_{n+1}} \subseteq W_{\tau_{\leq n}|-|}$: by Lemma 3.2 the local objects of $W_{|-|} \vee W_{\text{cosk}_{n+1}}$ are the constant objects on n -truncated objects, a reflective subcategory with reflector $c\tau_{\leq n}|-|$, so $\text{Sat}(W_{|-|} \vee W_{\text{cosk}_{n+1}}) = W_{\tau_{\leq n}|-|}$ ([7, Thm. 8.3], whose proof holds in $\mathcal{E}$), and $W_{\text{cosk}_{n+1}}$ is contained in it. Condition (a): $u_X \in W_{|-|}$, so it suffices that t_K lies in the join for every $K \in \mathcal{E}$. Consider

$$\begin{array}{ccc} cK & \xrightarrow{\in W_{\text{cosk}_{n+1}}} \text{cosk}_{n+1} cK & \xrightarrow{u \in W_{|-|}} c|\text{cosk}_{n+1} cK| \\ & \searrow t_K & \downarrow \simeq \\ & & c\tau_{\leq n} K \end{array}$$

where the top row h is the coskeleton unit followed by the unit of $|-| \dashv c$, and the vertical equivalence is c of Proposition 3.3. Since c is fully faithful,

$h = c(h')$ for $h' = |h|$ composed with $|c|\text{cosk}_{n+1}cK| \simeq |\text{cosk}_{n+1}cK|$, and h' is the realization of the coskeleton unit, which by Proposition 3.3 is the truncation map $K \rightarrow \tau_{\leq n}K$. So the triangle commutes and $t_K \simeq h$ is a composite of a $W_{\text{cosk}_{n+1}}$ -map, a $W_{|-|}$ -map and an equivalence, hence lies in the join. The join theorem gives $W_{|-|} \vee W_{\text{cosk}_{n+1}} = P^*(\text{Iso}) = W_{\tau_{\leq n}|-|}$; the local objects were identified above. $\square$

Remark 4.2. For $\mathcal{E} = \mathcal{S}$ and levelwise discrete objects the theorem recovers [6, Thm. 7.2]. There the proof used Kan replacement; here no fibrancy condition appears, because the coskeleton is applied to a constant object and Proposition 3.3 absorbs the Kan-complex input. Theorem 4.1 also shows that the zigzag needed to connect cK to $c\tau_{\leq n}K$ inside the join has length two, a $W_{\text{cosk}_{n+1}}$ -map followed by a $W_{|-|}$ -map.

5. THE THREE-FOLD JOIN

Let $L': \mathcal{E} \rightarrow \mathcal{E}$ be a left-exact localization (the reflector of a subtopos; in the language of [1], a lex modality), applied levelwise to $\mathcal{P}$, where it is again a lex reflector; write $W_{L'}$ for the class of maps f with $L'f_m$ an equivalence for all m .

Proposition 5.1. $W_{|-|} \vee W_{L'} = W_{L'|-|}$ and $W_{\text{cosk}_{n+1}} \vee W_{L'} = W_{\text{cosk}_{n+1}L'}$, the classes inverted by the reflectors $cL'|-|$ (onto constant objects on L' -local objects) and $\text{cosk}_{n+1}L'$ (onto $(n+1)$ -coskeletal levelwise L' -local objects).

Proof. Both are instances of [6, Cor. 4.2]: for two reflectors R_1, R_2 such that R_2R_1 inverts W_{R_2} , one has $W_{R_1} \vee W_{R_2} = W_{R_2R_1}$ (the join theorem with $P = R_2R_1$ and π the composite of the two units). For the first identity take $R_1 = L'$ (levelwise) and $R_2 = c|-|$: since L' preserves colimits, $|L'f| \simeq L'|f|$, so $cL'|-| = R_2R_1$ inverts $W_{L'}$, and the join is $W_{R_2R_1} = W_{L'|-|}$; its local objects are the constant objects on L' -local objects, the essential image of $cL'|-|$. For the second take $R_1 = L'$ and $R_2 = \text{cosk}_{n+1}$: $\text{cosk}_{n+1}L' \simeq L'\text{cosk}_{n+1}$ by Lemma 3.1, so R_2R_1 inverts W_{R_2} and the join is $W_{\text{cosk}_{n+1}L'}$, with local objects the $(n+1)$ -coskeletal levelwise L' -local objects. $\square$

Theorem 5.2. In $\text{Same}(\text{Fun}(\Delta^{\text{op}}, \mathcal{E}))$, for every $n \geq -1$,

$$W_{|-|} \vee W_{\text{cosk}_{n+1}} \vee W_{L'} = W_{L'\tau_{\leq n}|-|} = \{f : L'\tau_{\leq n}|f| \text{ an equivalence}\},$$

and the local objects of this class are the constant simplicial objects on n -truncated L' -local objects of $\mathcal{E}$.

Proof. By Theorem 4.1 the left side is $W_{\tau_{\leq n}|-|} \vee W_{L'}$. Apply the join theorem with $U = W_{\tau_{\leq n}|-|}$, $V = W_{L'}$, $W = \text{Iso}$, $P = cL'\tau_{\leq n}|-|$ and π_X the composite

$$X \xrightarrow{\in W_{\tau_{\leq n}|-|}} c\tau_{\leq n}|X| \xrightarrow{\in W_{L'}} cL'\tau_{\leq n}|X|,$$

the first map being $t_{|X|}u_X$ and the second the levelwise L' -unit of the constant object $c\tau_{\leq n}|X|$. This gives (a). For (b), P inverts U trivially; and if $f \in W_{L'}$ then $L'\tau_{\leq n}|f| \simeq \tau_{\leq n}L'|f| \simeq \tau_{\leq n}|L'f|$ is an equivalence, because L'

commutes with truncation and, being a left adjoint, with colimits. Hence the join is $P^*(\text{Iso}) = W_{L'\tau_{\leq n}|-}$. The local objects are those of $W_{\tau_{\leq n}|-}$ and of $W_{L'}$ simultaneously: constant objects on n -truncated objects that are also levelwise L' -local. $\square$

Corollary 5.3. $\bigcap_n (W_{|-} \vee W_{\text{cosk}_{n+1}} \vee W_{L'})$ is the class of maps f with $L'|f|$ ∞ -connective; if the subtopos of L' is hypercomplete, this is $W_{L'|-}$, the class of maps whose realization becomes an equivalence after L' -sheafification.

Proof. $L'\tau_{\leq n}|f| \simeq \tau_{\leq n}L'|f|$ is an equivalence for all n iff $L'|f|$ is ∞ -connective, and ∞ -connective maps of a hypercomplete ∞ -topos are equivalences. $\square$

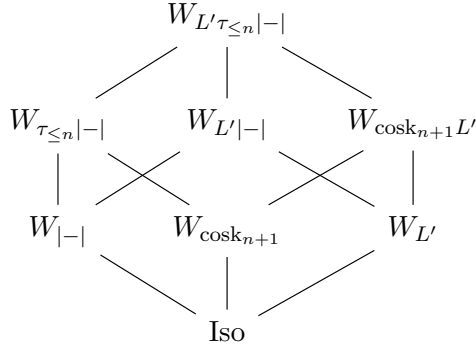


FIGURE 1. The sublattice of $\text{Same}(\text{Fun}(\Delta^{\text{op}}, \mathcal{E}))$ generated by the three classes: every join is the class of equivalences of a reflector (Theorem 4.1, Proposition 5.1, Theorem 5.2). Lines are joins, read upwards; incomparabilities are not asserted.

6. CONCLUSION

For simplicial objects in an ∞ -topos, the three canonical reflective localizations — onto constant objects, onto coskeletal objects, onto levelwise sheaves — have classes of equivalences whose joins in the lattice of samenesses are again classes of equivalences of reflectors, and the top join is the n -truncated L' -sheaf of the realization (Figure 1). The identities hold on the nose, which is stronger than the corresponding statement about local objects (equivalently, about saturations): two-out-of-three closure alone already reaches every map inverted by the composite reflector. The proof isolates one fact, Proposition 3.3: although the coskeleton of a constant simplicial object has highly non-truncated levels, its realization is the Postnikov section, and the coskeleton unit realizes to the truncation map. Its proof is a bisimplicial diagonal argument that converts the statement into the classical

one about Kan complexes, and it passes to every ∞ -topos because coskeleta are finite limits.

Two questions remain. First, whether the analogous identity holds for *cubical* objects in an ∞ -topos with the cubical coskeleta; for cubical sets it does [6], and the argument above reduces the general case to a cubical analogue of Proposition 3.3. Second, which other pairs of reflectors of $\mathrm{Fun}(\Delta^{\mathrm{op}}, \mathcal{E})$ have joins computed by a composite reflector; the coskeleta cosk_k and cosk_l themselves form a chain, and the interaction of cosk_{n+1} with the levelwise truncation $\tau_{\leq m}$ for $m \neq n$ is not treated here.

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